UDITYA NARAYAN TIWARI

AI/ML Engineer | Generative AI Developer | Open Source Co-author
+91 8169144110 | uditmerit@gmail.com | Uttar Pradesh, India
LinkedIn | GitHub | Portfolio | LeetCode

SUMMARY

AI/ML engineer focused on generative AI, NLP, RAG pipelines, semantic search, and REST APIs. Co-author of **RAGWire** and **SQL Notebook Studio**, with two additional Python libraries published on PyPI. Experienced with LangChain, FAISS, ChromaDB, FastAPI, Docker, and AWS; strong foundation in data structures and algorithms.

EDUCATION

VIT Bhopal University | B.Tech in Computer Science and Engineering (AI & ML) | 2023 - Present
Sehore, Madhya Pradesh | CGPA: **9.00 / 10.00**

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL
AI/ML: Machine learning, deep learning, NLP, feature engineering, model evaluation, predictive analytics
Generative AI: LLMs, RAG, embeddings, semantic search, vector databases
Frameworks: LangChain, LangGraph, Scikit-learn, TensorFlow, PyTorch, Transformers, Sentence Transformers, FastAPI, Flask, Streamlit
Data: Pandas, NumPy, Matplotlib, Plotly, FAISS, ChromaDB
Cloud & Tools: AWS, Docker, Git, GitHub, VS Code, Jupyter Notebook, Kaggle

OPEN SOURCE CO-AUTHORSHIP

RAGWire | Co-author | Python, LangChain, Qdrant | GitHub

- Co-authored a Python toolkit for multi-format document ingestion, metadata-aware retrieval, and grounded question answering, distributed on PyPI.
- Project capabilities include hybrid dense/sparse search, cross-encoder reranking, cited answers, SHA256 deduplication, and local/S3 source synchronization.

SQL Notebook Studio | Co-author | TypeScript, VS Code, SQL | GitHub

- Co-authored a VS Code extension providing Jupyter-style SQL notebooks for MySQL/MariaDB, PostgreSQL, SQL Server, SQLite, and DuckDB.
- Combines interactive result grids, charts, schema-aware autocomplete, SSH tunnels, and AI-assisted SQL generation, debugging, optimization, and explanation.

SELECTED PROJECTS

NeuroLens: Knowledge Retrieval Engine | GitHub

- Built a RAG platform for context-aware question answering across PDFs, documents, and web content using LLMs and LangChain.
- Implemented semantic retrieval with FAISS and ChromaDB and automated document ingestion, indexing, and retrieval workflows.

Smart Medical Care for Rural Areas | GitHub

- Developed a symptom-based medicine recommendation system using NLP, Sentence Transformers, cosine similarity, and Maximal Marginal Relevance (MMR).
- Built data preprocessing pipelines and Flask REST services for real-time recommendations.

MultiModel Analysis Package | Python, Scikit-learn, Pandas, NumPy | Documentation

- Published a PyPI ML benchmarking framework with 8,500+ downloads; supports 15+ classification and regression algorithms and reduces manual evaluation effort by 80%.
- Automated scaling, data splitting, exception handling, metrics, visualizations, and export to JSON and Excel; reports ROC-AUC, accuracy, F1, and R-squared.

LEADERSHIP & COMMUNITY

Microsoft Learn Student Chapter | Technical Team Member & Community Admin | Apr 2025 - Apr 2026

- Organized workshops, coding competitions, and hackathons focused on Microsoft technologies, cloud computing, and AI; coordinated student developers and technical knowledge-sharing sessions at VIT Bhopal University.

ACHIEVEMENTS & CERTIFICATIONS

- **Python package publishing:** Published NLP Text Preprocessing and MultiModel Analysis on PyPI, with 18,500+ combined downloads.
- **Microsoft Azure Data Fundamentals:** Certification covering cloud fundamentals, Azure services, security, governance, and business applications.
- **IIRS, ISRO:** Certification in AI/ML for Geodata Analysis, covering remote sensing and geospatial intelligence.